

Mean Squared Error

- The *mean squared error* (MSE) is a measure of the quality of an estimator.
- That is, the average squared difference between the estimated values and the true values:

$$\text{MSE}(\hat{\theta}) = E_{\theta} \left[(\hat{\theta} - \theta)^2 \right]$$

- Since this definition depends on the unknown parameter θ , the MSE is a prior property of an estimator.

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= \mathbf{E}_{\theta} \left[(\hat{\theta} - \theta)^2 \right] \\
&= \mathbf{E}_{\theta} \left[\left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] + \mathbf{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \right] \\
&= \mathbf{E}_{\theta} \left[\left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right)^2 + 2 \left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right) \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right) + \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \right] \\
&= \mathbf{E}_{\theta} \left[\left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right)^2 \right] + \mathbf{E}_{\theta} \left[2 \left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right) \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right) \right] + \mathbf{E}_{\theta} \left[\left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \right] \\
&\quad \text{constant} \qquad \qquad \qquad \text{constant} \\
&= \mathbf{E}_{\theta} \left[\left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right)^2 \right] + 2 \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right) \mathbf{E}_{\theta} \left[\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right] + \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \\
&\quad \text{constant} \\
&= \mathbf{E}_{\theta} \left[\left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right)^2 \right] + 2 \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right) \left(\mathbf{E}_{\theta}[\hat{\theta}] - \mathbf{E}_{\theta}[\hat{\theta}] \right) + \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \\
&\quad \text{zero} \\
&= \mathbf{E}_{\theta} \left[\left(\hat{\theta} - \mathbf{E}_{\theta}[\hat{\theta}] \right)^2 \right] + \left(\mathbf{E}_{\theta}[\hat{\theta}] - \theta \right)^2 \\
&= \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}_{\theta}(\hat{\theta}, \theta)^2
\end{aligned}$$

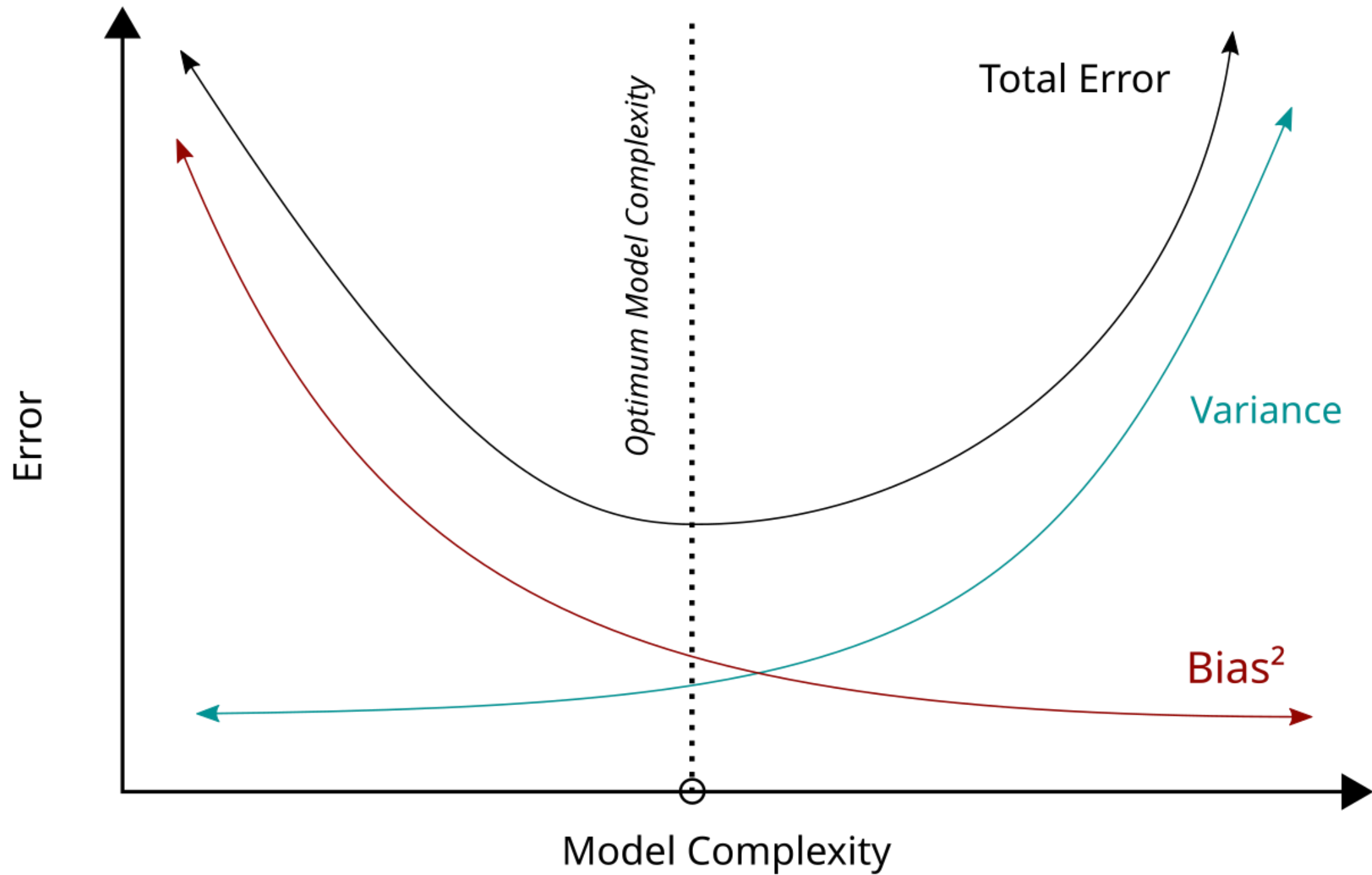
Shorter Proof

- It is well-known that for any random variable X , $E[X^2] = \text{Var}(X) + (E[X])^2$.

- By substituting X with $\hat{\theta} - \theta$, we have

$$\begin{aligned}\text{MSE}(\hat{\theta}) &= E_{\theta} [(\hat{\theta} - \theta)^2] \\ &= \text{Var}(\hat{\theta} - \theta) + E_{\theta}[\hat{\theta} - \theta]^2 \\ &= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2\end{aligned}$$

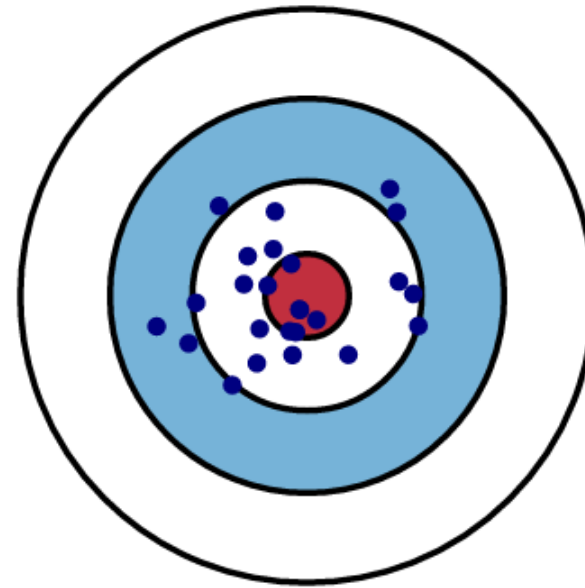
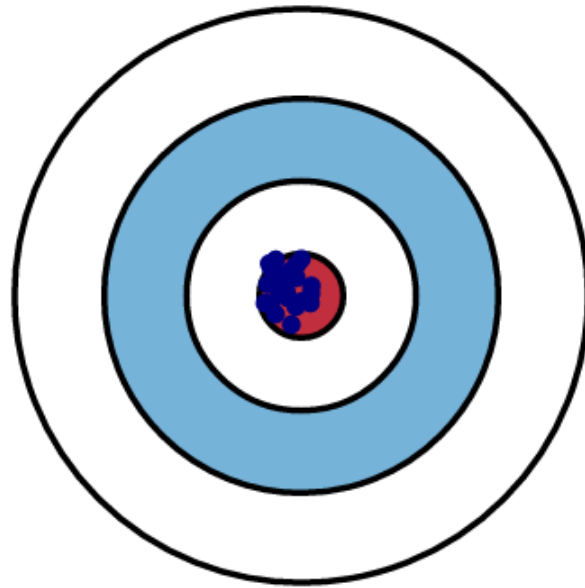
- For an unbiased $\hat{\theta}$, $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta})$.



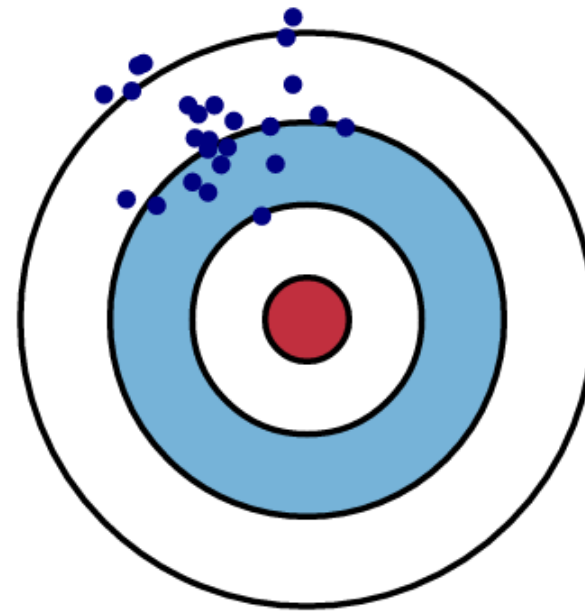
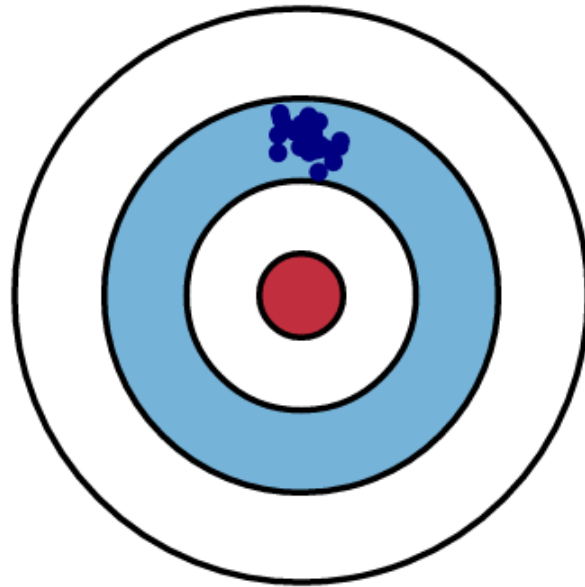
Low Variance

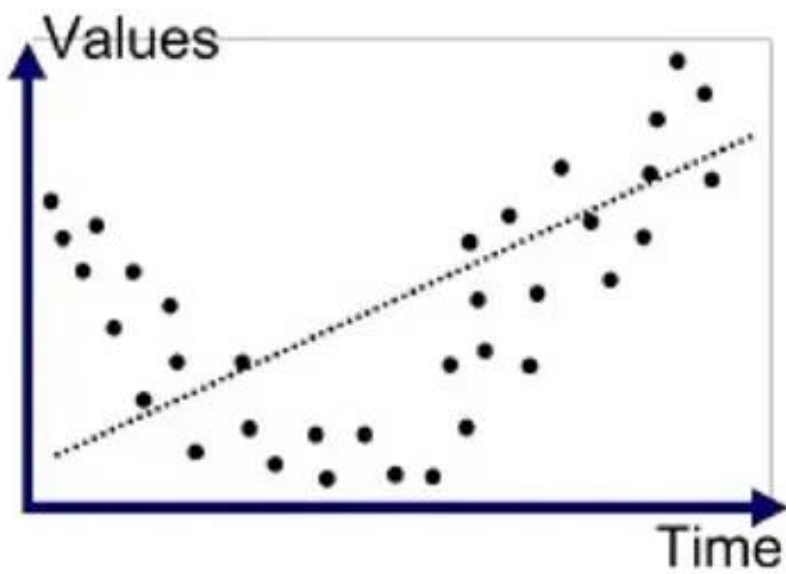
High Variance

Low Bias

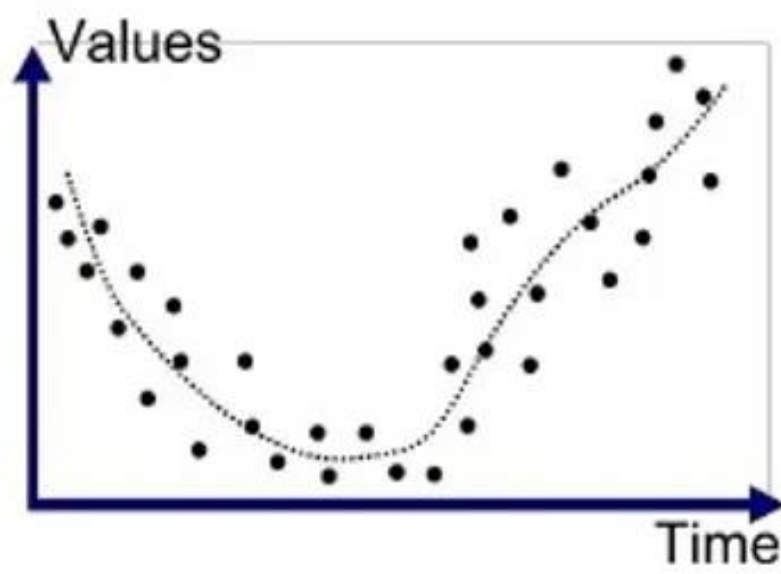


High Bias

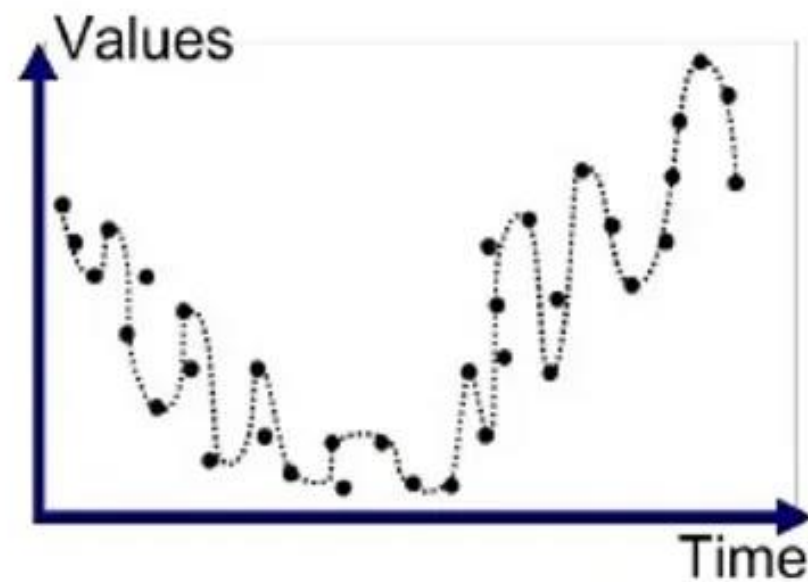




Underfitted



Good Fit/Robust



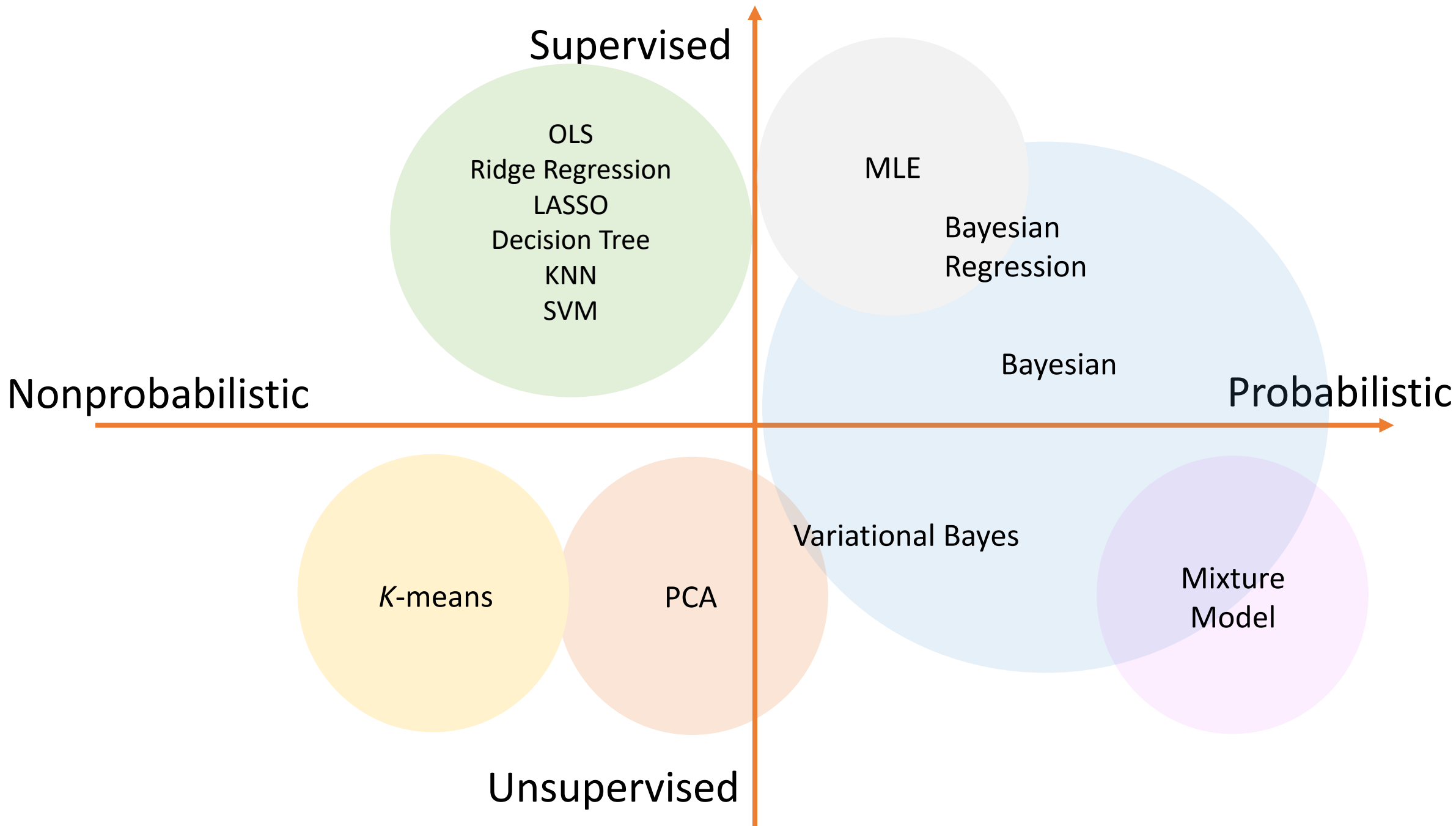
Overfitted

A First Course in Machine Learning

Chapter 3

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Maximum a Posterior (MAP)

- Now we know the posterior is

$$p(\mathbf{w}|\mathbf{t}, \mathbf{X}, \sigma^2) = N(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w)$$
$$\boldsymbol{\mu}_w = \boldsymbol{\Sigma}_w \left(\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{t} + \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0 \right)$$
$$\boldsymbol{\Sigma}_w = \left(\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X} + \boldsymbol{\Sigma}_0^{-1} \right)^{-1}$$

- Even $\boldsymbol{\mu}_0 = \mathbf{0}$, $\boldsymbol{\mu}_w \neq \mathbf{b}_{\text{OLS}}$. $\mathbf{b}_{\text{MLE}}$ is independent of σ^2 . Since $\boldsymbol{\Sigma}_0$ is positive semi-definite, $\boldsymbol{\Sigma}_w$ is usually smaller than $\text{Var}(\mathbf{b}_{\text{OLS}})$.

Bayesian Predictions

Expectations?

Marginal?

Conditional

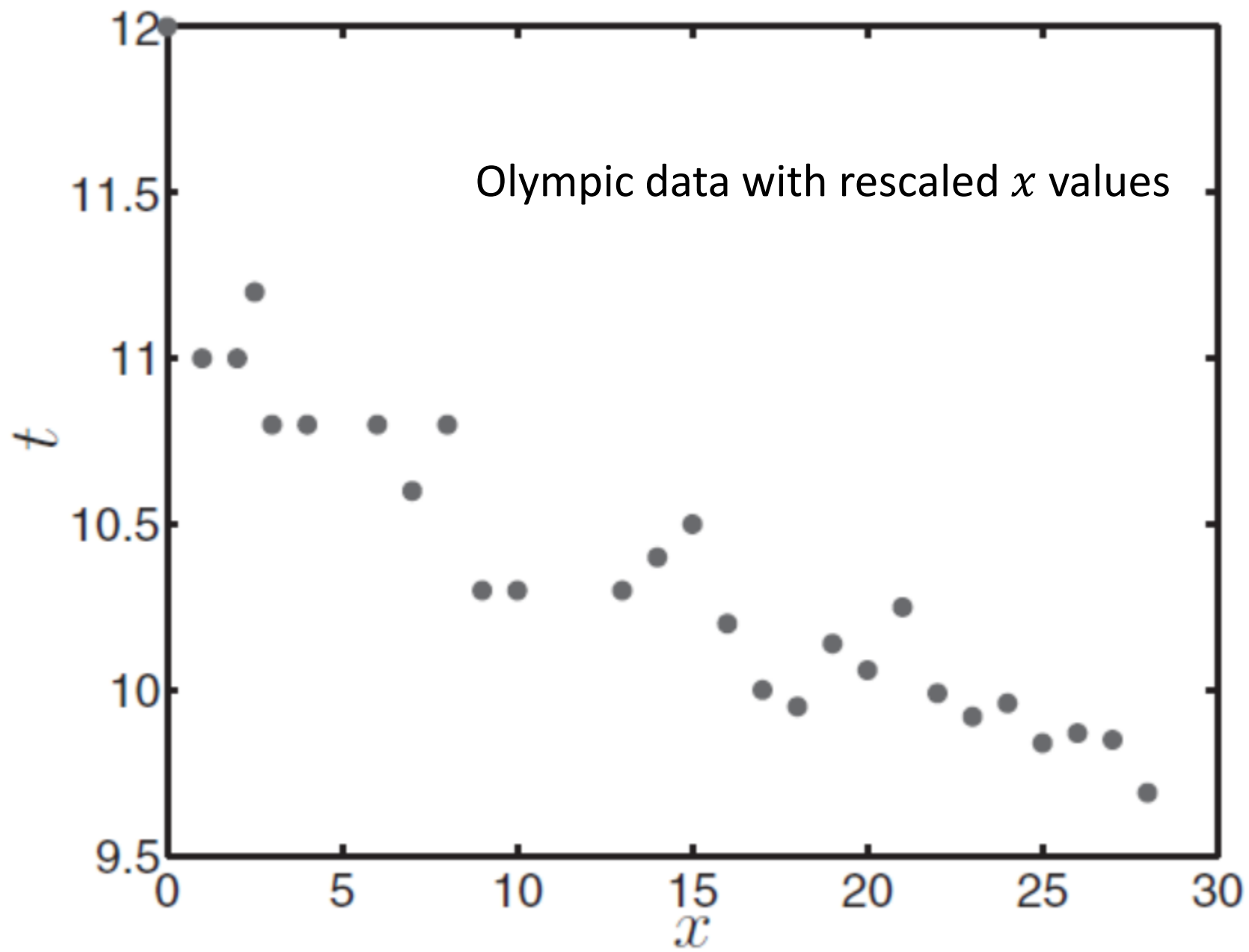
Probability?

- Given a new observation $\mathbf{x}_{new}$, predictions are

→
$$p(t_{new}|\mathbf{x}_{new}, \mathbf{X}, \mathbf{t}, \sigma^2) = \mathbf{E}_{\mathbf{w}}\{p(t_{new}|\mathbf{x}_{new}, \mathbf{w}, \sigma^2)\}$$
$$= \int p(t_{new}|\mathbf{x}_{new}, \mathbf{w}, \sigma^2)p(\mathbf{w}|\mathbf{X}, \mathbf{t}, \sigma^2) d\mathbf{w}$$

- We define that $p(t_{new}|\mathbf{x}_{new}, \mathbf{w}, \sigma^2) = N(\mathbf{x}_{new}^T \mathbf{w}, \sigma^2)$, and we just solve that $p(\mathbf{w}|\mathbf{X}, \mathbf{t}, \sigma^2) = N(\mu_{\mathbf{w}}, \Sigma_{\mathbf{w}})$, predictions must follow a normal distribution, i.e.,

$$p(t_{new}|\mathbf{x}_{new}, \mathbf{X}, \mathbf{t}, \sigma^2) = N(\mu_{new}, \sigma_{new})$$

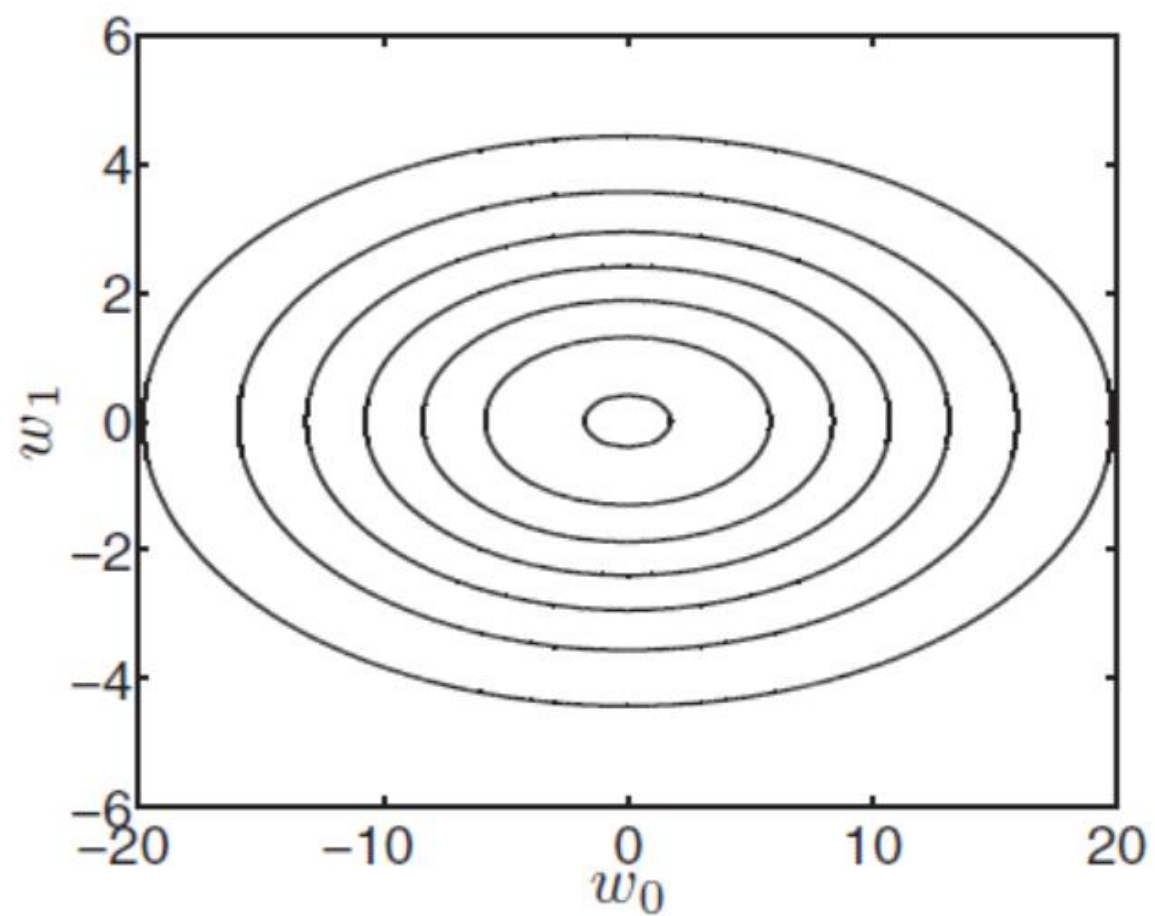


Gaussian Prior

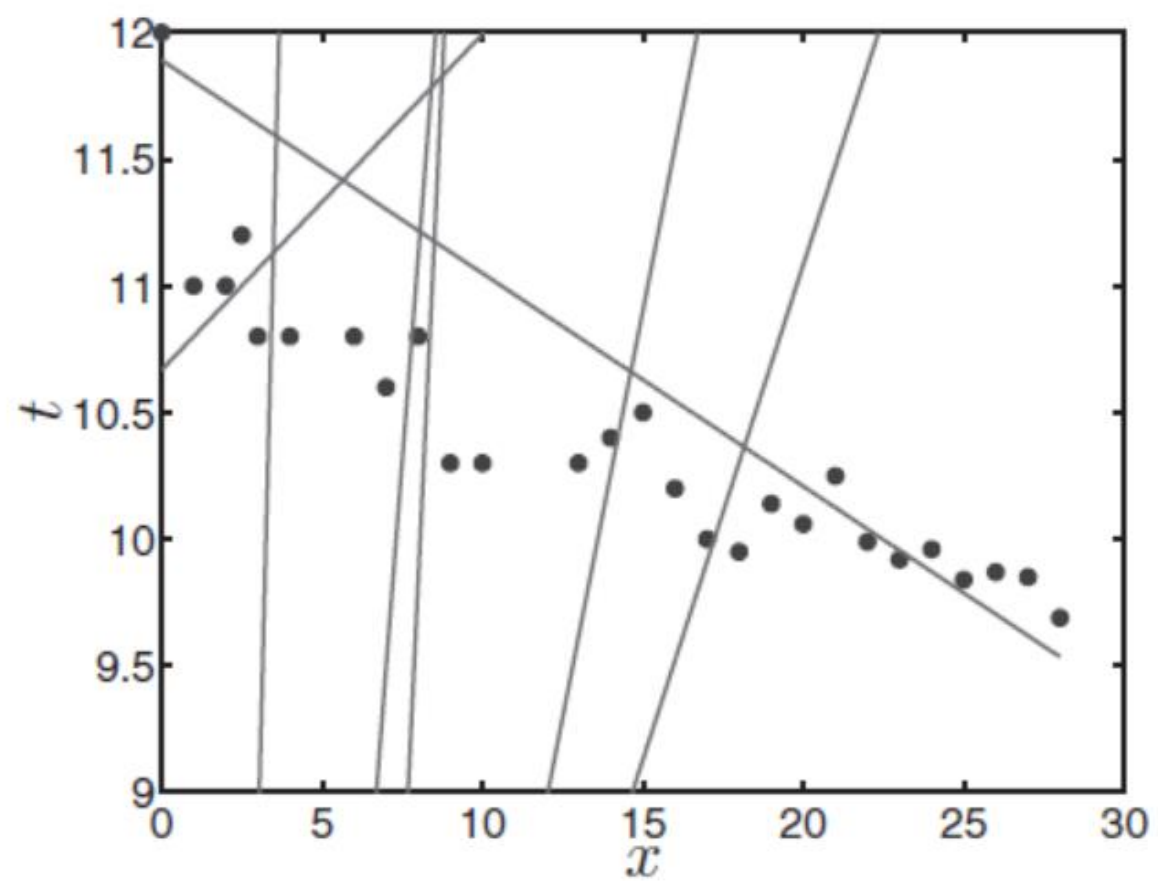
- We illustrate the prior and posterior with a first-order polynomial. So the prior $p(\mathbf{w}|\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) = N(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ where $\boldsymbol{\mu}_0 = [0,0]^T$ and

$$\boldsymbol{\Sigma}_0 = \begin{bmatrix} 100 & 0 \\ 0 & 5 \end{bmatrix}$$

- We assume the two parameters are independent. Since the MLE of w_0 is much larger than w_1 , the larger value for variance is assigned to w_0 .
- Whether w_0 and w_1 are independent in the posterior?



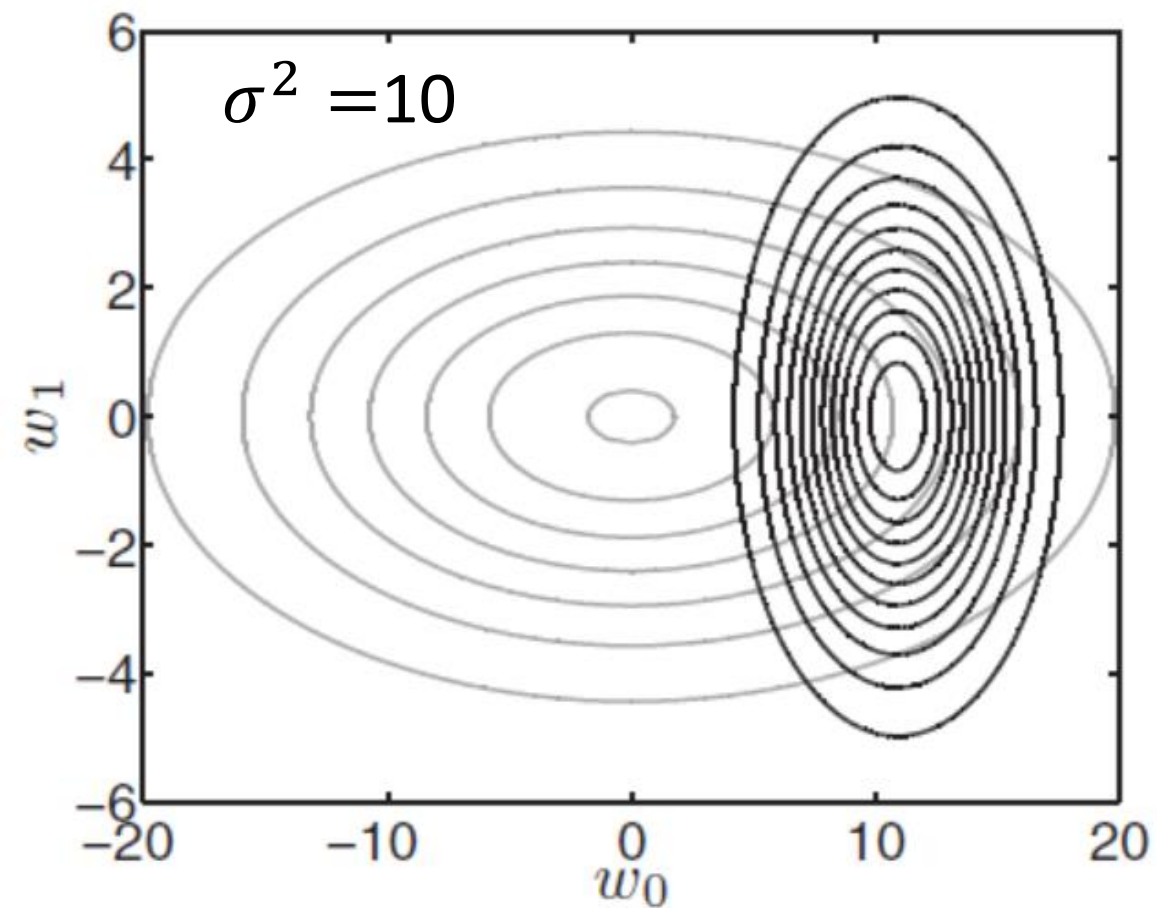
(a) Prior density.



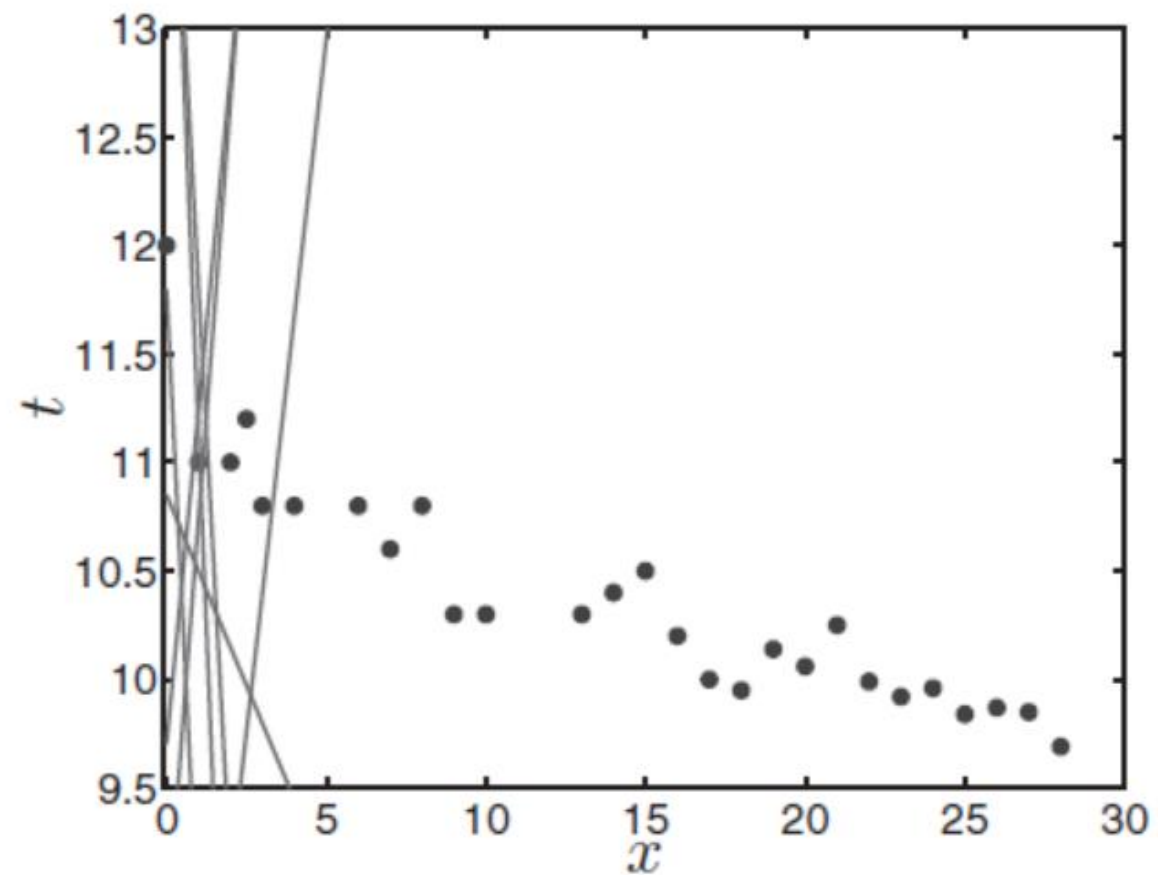
(b) Functions created from parameters drawn from prior.

Gaussian Likelihood

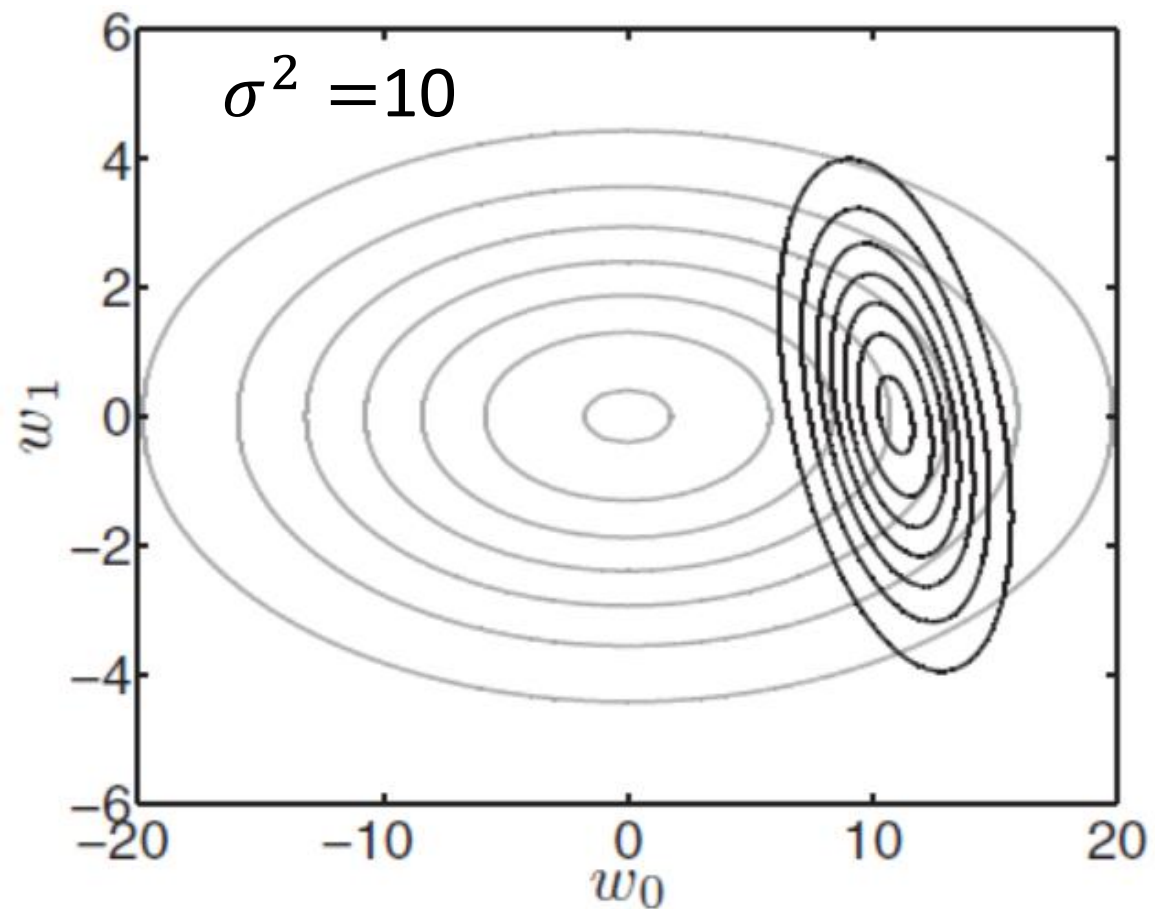
- Given the prior $p(\mathbf{w}|\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) = N(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and the likelihood $p(\mathbf{t}|\mathbf{w}, \mathbf{X}, \sigma^2) = N(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N)$, we can compute the posterior after we observe several data points.
- Assume that we know the true value of σ^2 .
- The posterior distribution must be a normal distribution. The more data points considered, the more condensed the posterior is--and *the more tilted it is!*



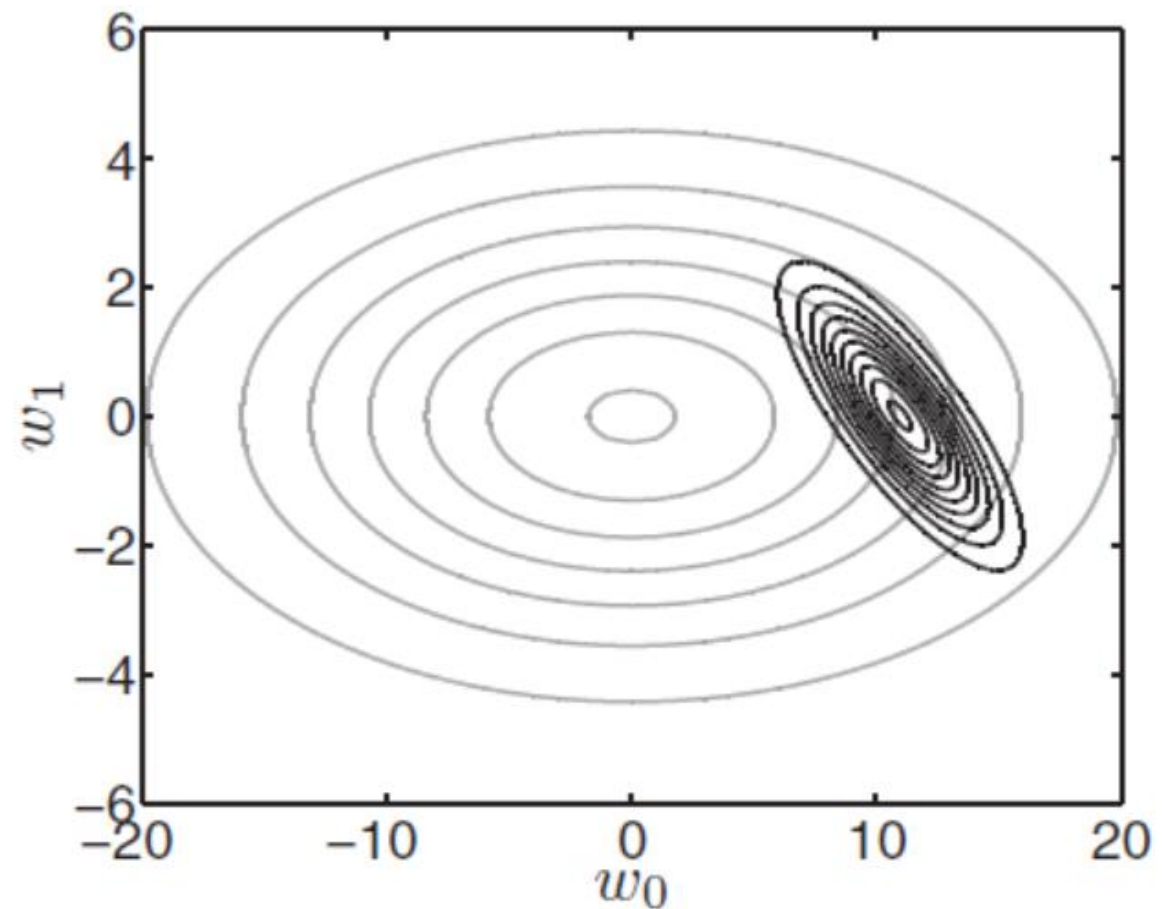
(a) Posterior density (dark contours) after the first data point has been observed. The lighter contours show the prior density.



(b) Functions created from parameters drawn from the posterior after observing the first data point.



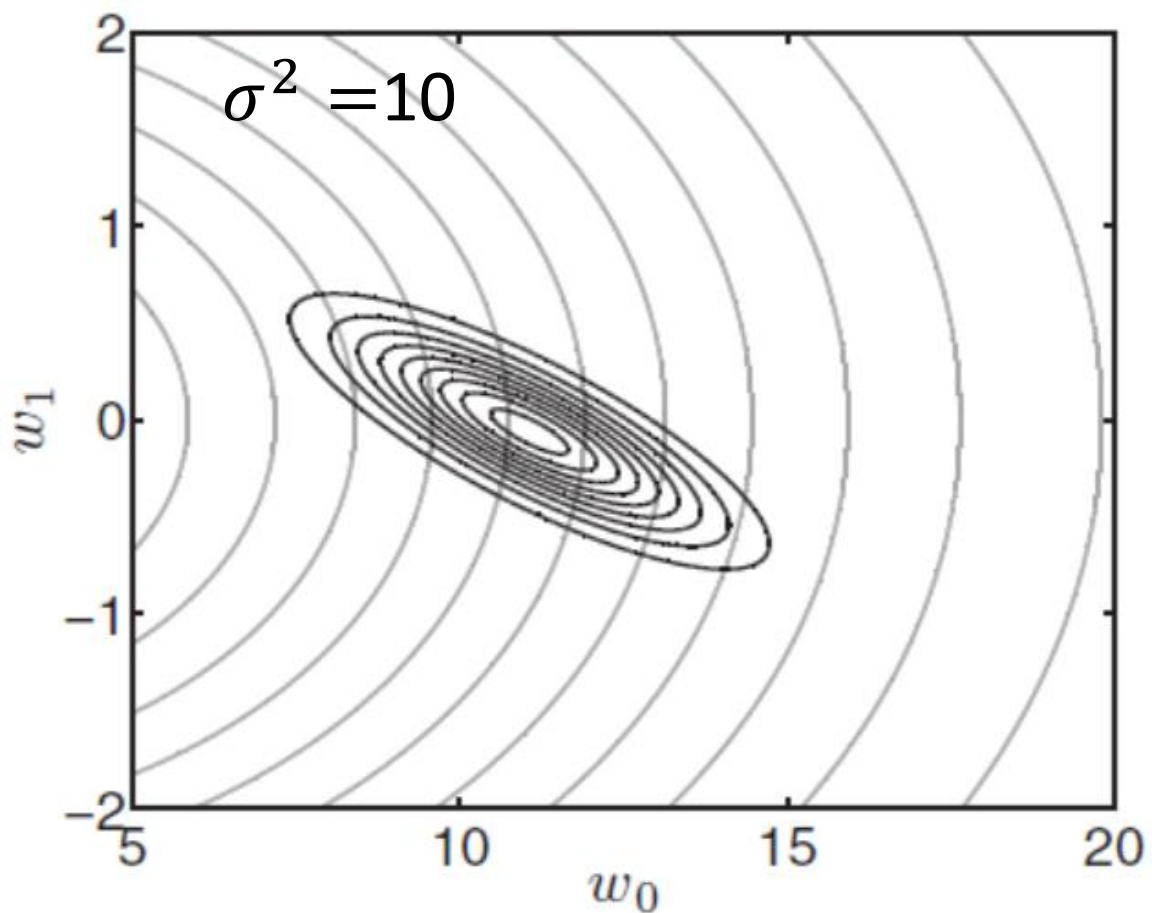
(c) Posterior density (dark contours) after the first two data points have been observed.



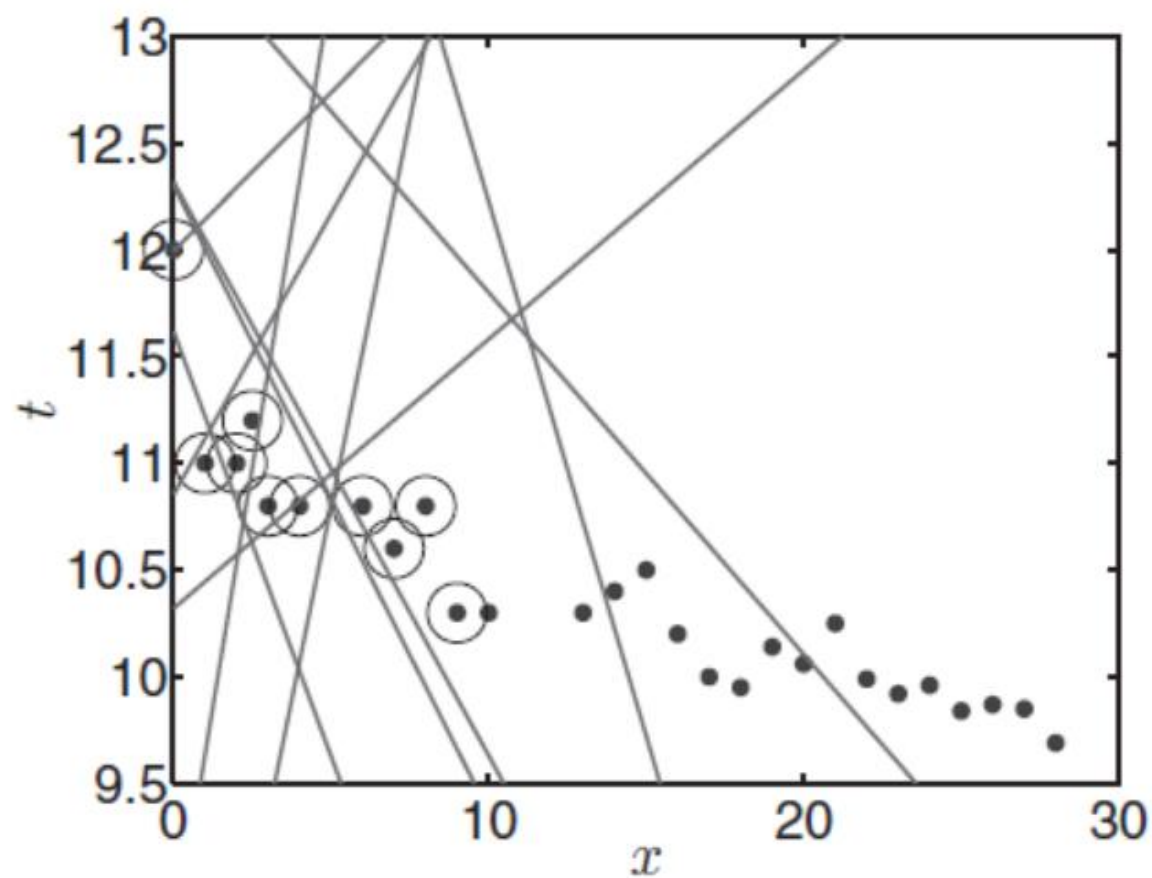
(d) Posterior density (dark contours) after the first five data points have been observed.

Evolution of Posterior

- The posterior begins to tilt indicating a dependence between two parameters.
- This dependence comes entirely from the evidence within the (say first ten) data points.
- When compared with Fig 3.20(b), the posterior begins to favor parameters that correspond to models suited to our data.



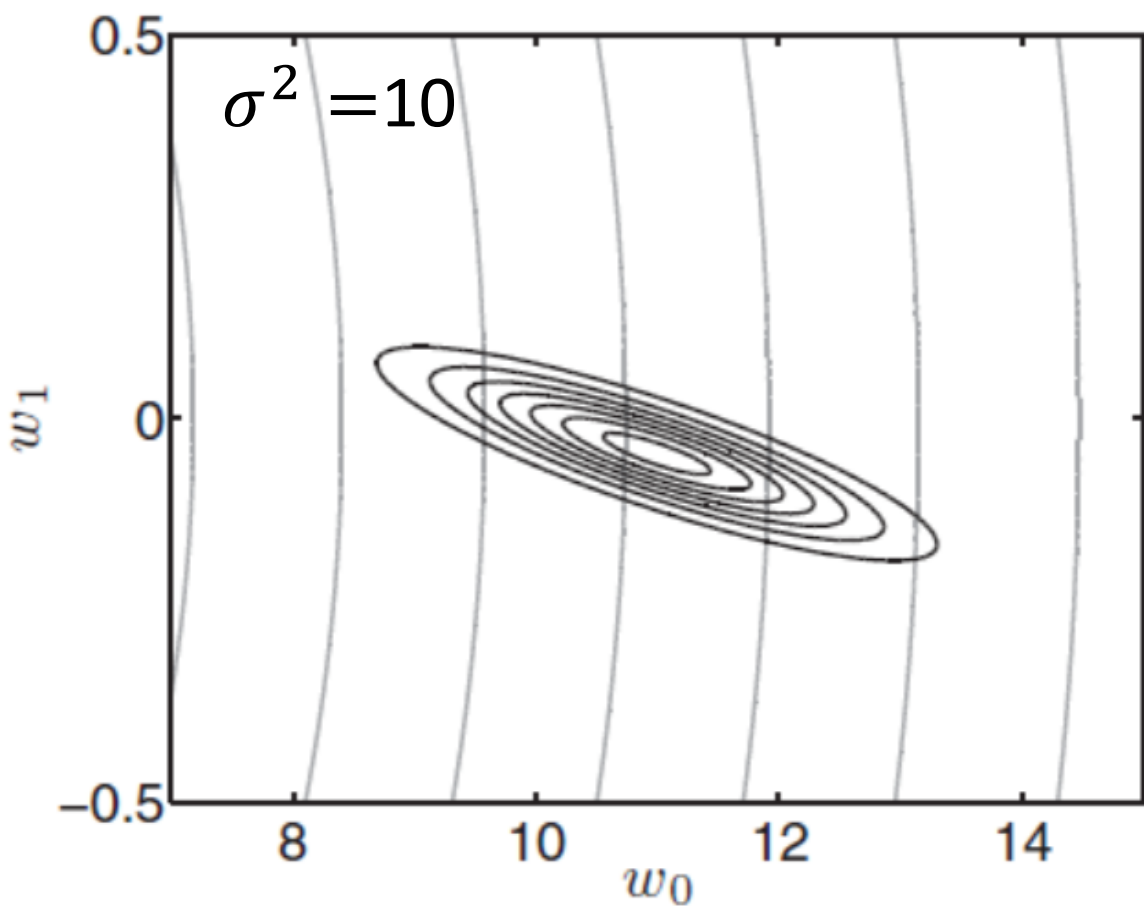
(e) Posterior density (dark contours) after the first ten data points have been observed. (Note that we have zoomed in.)



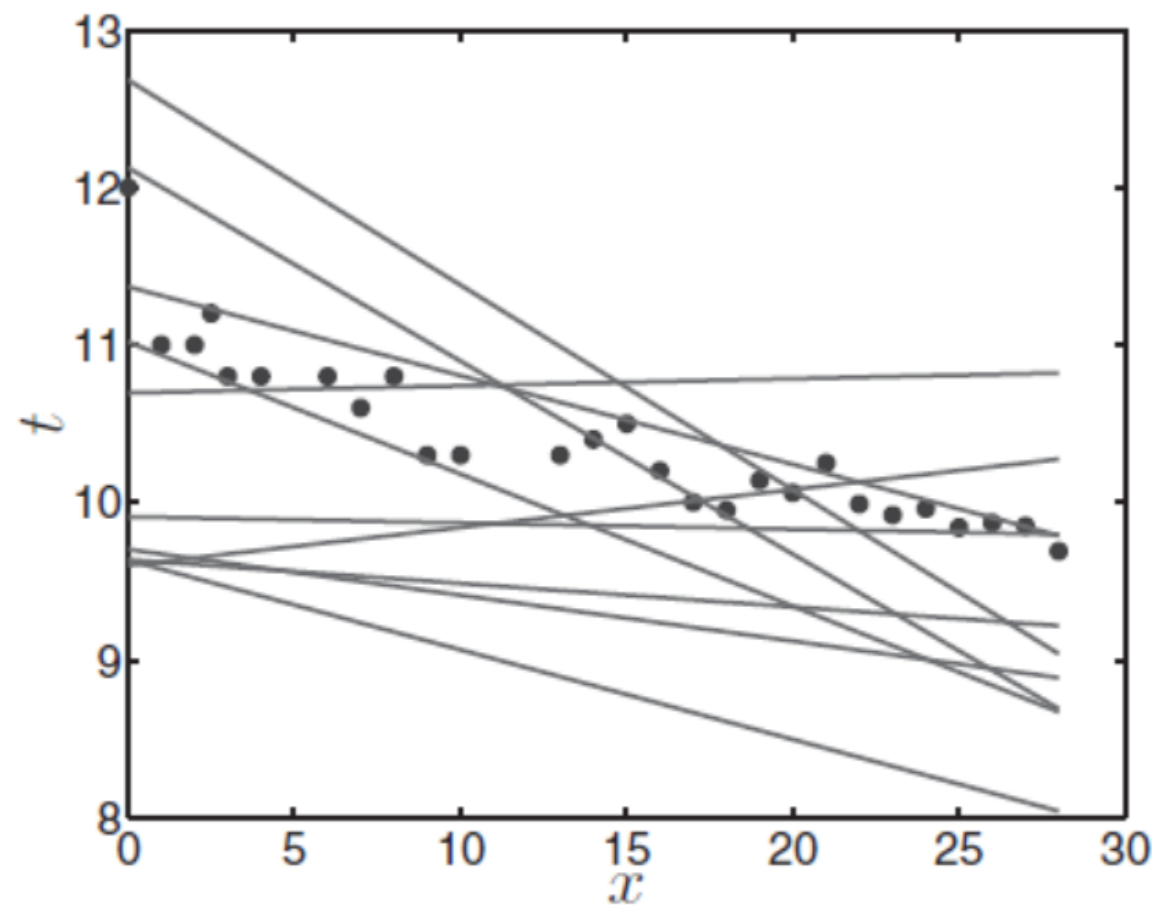
(f) Functions created from parameters drawn from the posterior after observing the first ten (highlighted) data points.

Final Posterior

- After all 27 data points are included, we can see functions drawn from the final posterior.
- The functions are really now beginning to follow the trend in our data.
- There is still a lot of variability though, due to the relatively high value of $\sigma^2 = 10$ (to help visualization).



(a) Posterior density (dark contours) after all datapoints have been observed. The lighter contours show the prior density. (Note that we have zoomed in.)



(b) Functions created from parameters drawn from the posterior after observing all data points.